

CV 2080 — STRUCTURAL DESIGN

JAN – MAY 2026

TERM PROJECT · DELIVERABLE 1

Analysis and Design of a G+2 Storey Building

*Preliminary Assumptions · Methodology · Member Sizing
Load Calculations · Structural Model*

TEAM NAME	GROUP 8
COURSE	CV 2080 - Structural Design
BATCH	ODD
OCCUPANCY TYPE	Residential

TEAM MEMBERS

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GROUP 8 · CV 2080 · 16th MARCH 2026

■ 1. Introduction & Building Configuration

This report constitutes Deliverable 1 of the CV 2080 Term Project. The objective of this project is to analyse and design a G+2 storey Residential building for both a Reinforced Concrete (RC) frame and a Structural Steel frame, considering gravity loads only (dead load + live load).

This deliverable documents: (i) all assumptions and methodology to be adopted; (ii) preliminary member cross-section sizes; (iii) complete load calculations; (iv) ETABS analysis results, including BMDs, SFDs, deflection profiles, member force summaries, and support reaction verification; and (v) a description of both structural models.

BUILDING TYPE	G+2 (Ground + 2 upper floors, 3 storeys total)
BATCH	ODD
OCCUPANCY	Residential
STOREY HEIGHT	5.5 m (uniform all floors)
PLAN DIMENSION	9 m × 9 m (square)
BAYS IN X-DIRECTION	2 equal bays @ 4.5 m each
BAYS IN Y-DIRECTION	3 equal bays @ 3.0 m each
COLUMN GRID	3 columns × 4 columns = 12 columns per floor
TOTAL BUILDING HEIGHT	3 × 5.5 = 16.5 m
FRAME TYPES	RC Frame (IS 456) and Steel Frame (IS 800)
LOADING CONSIDERED	Gravity only — Dead Load (DL) + Live Load (LL)

■ 2. Assumptions

The following assumptions have been made. All are in accordance with standard Indian codes of practice and are clearly stated for transparency.

◆ 2.1 Structural System

- The structure is a 3D moment-resisting frame (MRF) with no shear walls.
- All columns are fixed at the base (rigid foundation assumed — detailed foundation design is out of scope).
- Floors consist of two-way RC slabs supported on primary beams running in both X and Y directions.
- No secondary beams are assumed at this stage; slab panels are directly supported on primary beams.
- Infill walls are not modelled; their loads are approximated as equivalent distributed loads.

◆ 2.2 Materials — RC Frame

PROPERTY	VALUE
Concrete grade	M25 ($f_{ck} = 25 \text{ N/mm}^2$)
Steel reinforcement	Fe 500 ($f_y = 500 \text{ N/mm}^2$) — HYSD Grade 500
Unit weight of concrete	25 kN/m^3
Modulus of elasticity of concrete	$E_c = 5000\sqrt{f_{ck}} = 25,000 \text{ N/mm}^2$
Poisson's ratio	0.2
Clear cover - beams	25 mm (IS 456 Cl. 26.4.1)
Clear cover - columns	40 mm (IS 456 Cl. 26.4.2)

◆ 2.3 Materials — Steel Frame

PROPERTY	VALUE
Structural steel grade	Fe 250 per IS 2062
Yield strength	$f_y = 250 \text{ N/mm}^2$
Ultimate tensile strength	$f_u = 410 \text{ N/mm}^2$
Unit weight of steel	78.5 kN/m^3
Modulus of elasticity	$E_s = 2,00,000 \text{ N/mm}^2$
Floor slab	M25 slab (same for both models)
Beam-column connections	Rigid moment connections are assumed

◆ 2.4 Geometric Assumptions

- Slab panels are assumed to be two-way slabs with a short span-to-long span ratio ≤ 2.0 .
- Beam and column centre-line dimensions used in ETABS (finite joint offsets not applied at preliminary stage).
- Uniform storey height of 5.5 m for all floors, including the ground floor (unusually high — column sizes adjusted accordingly).

◆ 2.5 Analysis Assumptions

- Linear elastic first-order static analysis is used (geometric non-linearity neglected).
- Rigid floor diaphragm assumption is used in ETABS (in-plane floor stiffness is infinite).
- Self-weight of all members is automatically computed by ETABS based on unit weights and cross-section dimensions.
- $P-\Delta$ effects are not considered (gravity-only analysis, no lateral load).

◆ 2.6 Design Code References

- IS 456:2000 — Reinforced Concrete Design
- IS 800:2007 — Steel Structure Design
- IS 875 (Part 1):1987 — Dead Loads
- IS 875 (Part 2):1987 — Imposed Loads
- IS 875 (Part 5):1987 — Load Combinations
- SP 6(1):1964 — Structural Steel Sections Handbook

■ 3. Building Geometry and Configuration

◆ 3.1 Column Grid

The building has a regular 3×4 column grid. Grid lines are labelled A-B-C along X and 1-2-3-4 along Y. Column designations follow the format C-[X][Y] (e.g., C-A1 = corner column at X=1, Y=A).

COLUMN CATEGORY	GRID POSITIONS	COUNT PER FLOOR	NOTES
Corner columns	C-A1, C-C1, C-A4, C-C4	4	Tributary area ≈ 6.75 m ²
X-edge columns	C-B1, C-B4	2	Tributary area ≈ 13.5 m ²
Y-edge columns	C-A2, C-C2, C-A3, C-C3	4	Tributary area ≈ 6.75 m ²
Interior columns	C-B2, C-B3	2	Tributary area ≈ 13.5 m ²
Total per floor		12	

◆ 3.2 Beam Layout

- Primary beams run in both X and Y directions at every grid line on every floor.
- Total beams = $(2 \times 4 + 3 \times 3) \times 3 = 51$
- Slab panels: $2 \times 3 = 6$ panels per floor, each 4.5 m × 3.0 m.

■ 4. Preliminary Member Sizes

Preliminary dimensions are estimated using standard span/depth ratios and tributary area methods prior to analysis. These will be confirmed against ETABS outputs and refined in Deliverables 2 and 3.

◆ 4.1 Slab Thickness

For a two-way slab with longer span = 4.5 m, using IS 456 Cl. 24.1 (deflection control):

- Basic L/d ratio for two-way slab (simply supported) = 28
- Required effective depth = $4500 / 28 \approx 161$ mm $\rightarrow$ Overall depth ≈ 189 mm
- Adopted slab thickness: 150 mm (standard value per preliminary thumb rule)

◆ 4.2 RC Beam Sizes

Depth estimated from span/depth ratio of ~ 10 for primary beams carrying slab loads (based on the 5.5 m storey height, a slightly more conservative ratio is used):

BEAM ID	DIRECTI ON	SPAN (M)	ADOPTED D (MM)	WIDTH B (MM)	L/D
B-X-int & B-X-ext (X-beams)	X	4.5	450	230	$4500/450 = 10$ ✓
B-Y-int & B-Y-ext (Y-beams)	Y	3.0	450	230	Std. uniform depth

Note: A uniform beam size of 230 mm $\times$ 450 mm is adopted for all beams in both directions. This is conservative for the 3.0 m Y-direction span but simplifies construction and improves uniformity.

◆ 4.3 RC Column Sizes — Tributary Area Method with 1.5(DL+LL)

Factored load intensity = $1.5 \times (DL + LL) = 1.5 \times (5.5 + 2.0) = 11.25$ kN/m² per floor. Number of floors = 3. The 5.5 m storey height requires a 300 $\times$ 450 mm column to prevent slenderness/buckling issues.

COLUMN TYPE	TRIBUTARY AREA (M ²)	TOTAL PU (KN)	REQUIRED AG (MM ²)	ADOPTED SIZE
Interior (C-B2, C-B3)	$4.5 \times 3.0 = 13.5$	$11.25 \times 13.5 \times 3 = 456 \text{ kN}$	~54,000	300 × 450 mm
X-edge (C-B1, C-B4)	$4.5 \times 1.5 = 6.75$	$11.25 \times 6.75 \times 3 = 228 \text{ kN}$	~27,000	300 × 450 mm
Y-edge (C-A2 etc.)	$2.25 \times 3.0 = 6.75$	$11.25 \times 6.75 \times 3 = 228 \text{ kN}$	~27,000	300 × 450 mm
Corner (C-A1 etc.)	$2.25 \times 1.5 = 3.375$	$11.25 \times 3.375 \times 3 = 114 \text{ kN}$	~13,500	300 × 450 mm

Capacity check (interior column, $P_u = 456 \text{ kN}$): IS 456 Cl. 39.3 — $P_{u_cap} = 0.4 \times f_{ck} \times A_c + 0.67 \times f_y \times A_{sc}$. For 300×450 ($A_g = 135,000 \text{ mm}^2$) with 1.5% steel ($A_{sc} = 2,025 \text{ mm}^2$): $P_{u_cap} = 0.4 \times 25 \times 132,975 + 0.67 \times 500 \times 2,025 = 1,330 + 679 = 2,009 \text{ kN} \gg 456 \text{ kN}$. Adequate with margin for eccentricity.

A uniform 300×450 mm section is adopted for all columns to provide robustness against the 5.5 m storey height and to prevent slenderness buckling.

◆ 4.4 Steel Frame — Section Sizes

MEMBER	SPAN/TYPE	IS SECTION	IXX (CM ⁴)	WEIGHT (KG/M)	BASIS
X-dir Beams	4.5 m span	ISMB 300	8,603	44.2	$L/d \approx 4500/300 = 15 \checkmark$
Y-dir Beams	3.0 m span	ISMB 300	8,603	44.2	Uniform section
Int. Columns	H-section	ISHB 250	7,983	51.0	Axial load $\approx 608 \text{ kN}$
Edge Columns	H-section	ISHB 250	7,983	51.0	Axial load $\approx 304 \text{ kN}$
Corner Col.	H-section	ISHB 250	7,983	51.0	Axial load $\approx 152 \text{ kN}$

■ 5. Load Calculation and Application

All loads are calculated in accordance with IS 875 Parts 1 and 2. Loads are applied in ETABS as area loads on slabs and as line loads directly on perimeter beams for wall weights.

◆ 5.1 Dead Loads (Permanent Actions)

◆ 5.1.1 Floor Slab Dead Load

COMPONENT	CALCULATION	LOAD (KN/M ²)
RC Slab self-weight	$0.150 \text{ m} \times 25 \text{ kN/m}^3$	3.750
Floor finish (tiles + screed)	Assumed (IS 875 Pt.1)	1.000
False ceiling / MEP services	Assumed nominal	0.500
TOTAL FLOOR SUPERIMPOSED DL		$5.25 \approx 5.5 \text{ kN/m}^2$

◆ 5.1.2 Roof Dead Load

COMPONENT	CALCULATION	LOAD (KN/M ²)
RC Slab self-weight	0.150×25	3.750
Waterproofing treatment	Brick bat coba 75 mm	1.500
Roof finish (screed)	Assumed	0.500
TOTAL ROOF DL		$5.75 \approx 6.0 \text{ kN/m}^2$

◆ 5.2 Live Loads — IS 875 Part 2

AREA / OCCUPANCY	IS 875 PT. 2 REFERENCE	LIVE LOAD (KN/M ²)	APPLIED IN MODEL
Dwelling units / Residential floors	Table 1, Sl. 1	2.0	Floors 1, 2
Corridors/passages	Table 1, Sl. 4	3.0	Applied as 2.0 (conservative uniform for residential)
Accessible roof (flat roof)	Table 1, Sl. 1	1.5	Roof level

◆ 5.3 Load Combinations — IS 875 Part 5

COMBINATION	EXPRESSION	PURPOSE	APPLICABLE CODE
COMB 1 (Design)	1.5 (DL + LL)	Limit state strength design	IS 456 / IS 800

◆ 5.4 Total Applied Load Summary — Factored: 1.5(DL + LL)

All loads are computed using COMB 1 = 1.5(DL + LL). Factored intensities: Floor = $1.5 \times (5.5 + 2.0) = 11.25 \text{ kN/m}^2$; Roof = $1.5 \times (6.0 + 1.5) = 11.25 \text{ kN/m}^2$; Wall = $1.5 \times 19 = 28.5 \text{ kN/m}$.

LOAD SOURCE	CALCULATION	UNFACTORED (KN)	FACTOR	FACTORED (KN)
Floor DL × 2 floors	$5.5 \times 81 \times 2$	891.0	1.5	1,336.5
1.5 × Roof DL	$1.5 \times 6.0 \times 81$	729.0	1.5	729.0
Wall DL (perimeter, 3 floors)	$19 \times 36 \times 3$	2,052.0	1.5	3,078.0
Floor LL × 2 floors	$2.0 \times 81 \times 2$	324.0	1.5	486.0
1.5 × Roof LL	$1.5 \times 1.5 \times 81$	182.3	1.5	182.3
TOTAL (excl. self-weight)		3,874.5 kN	—	5,811.8 kN

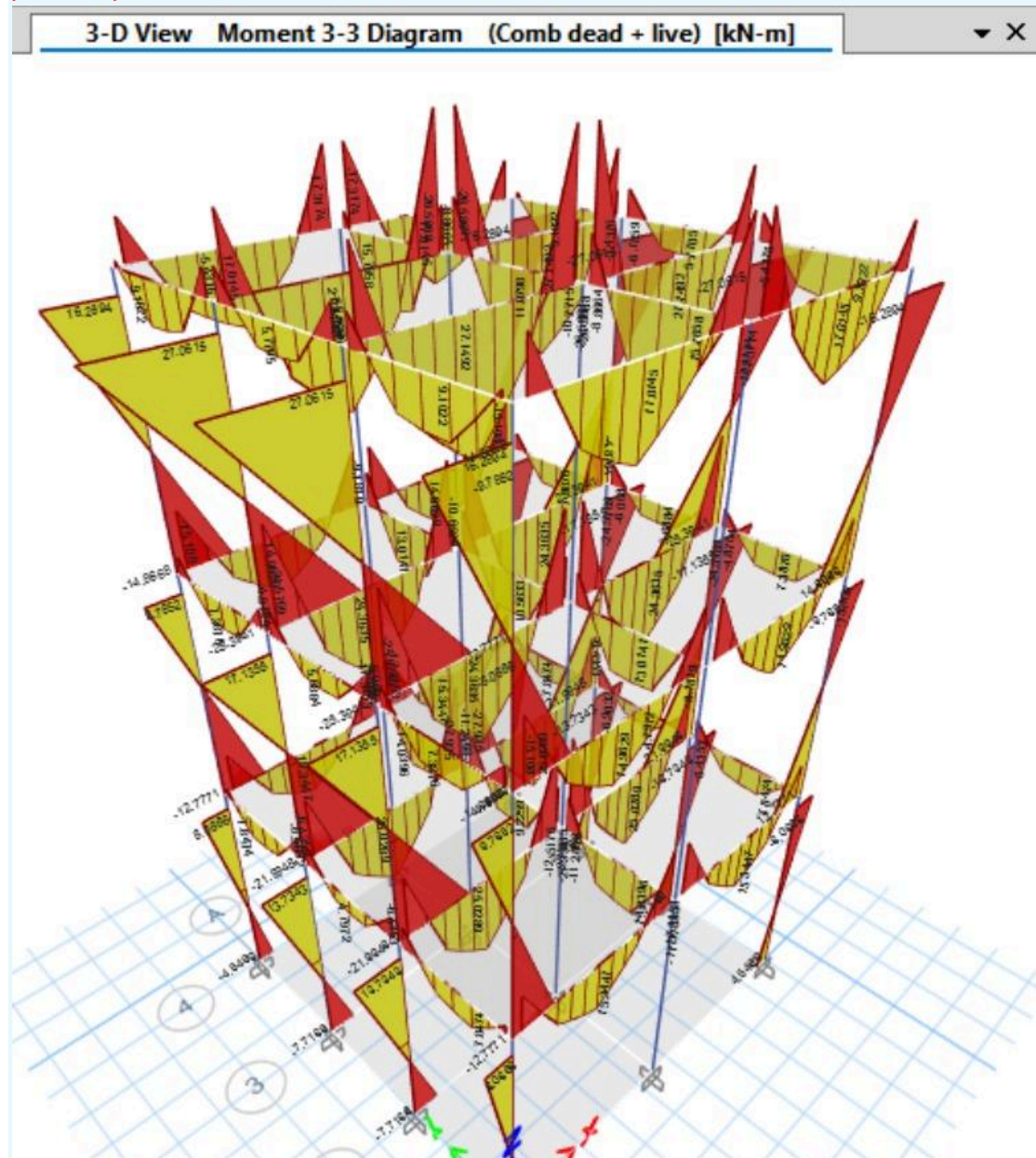
◆ **Note:** Self-weight of beams and columns is auto-computed by ETABS and included under the Dead load pattern. The factored total above is used to verify support reactions in Section 10.

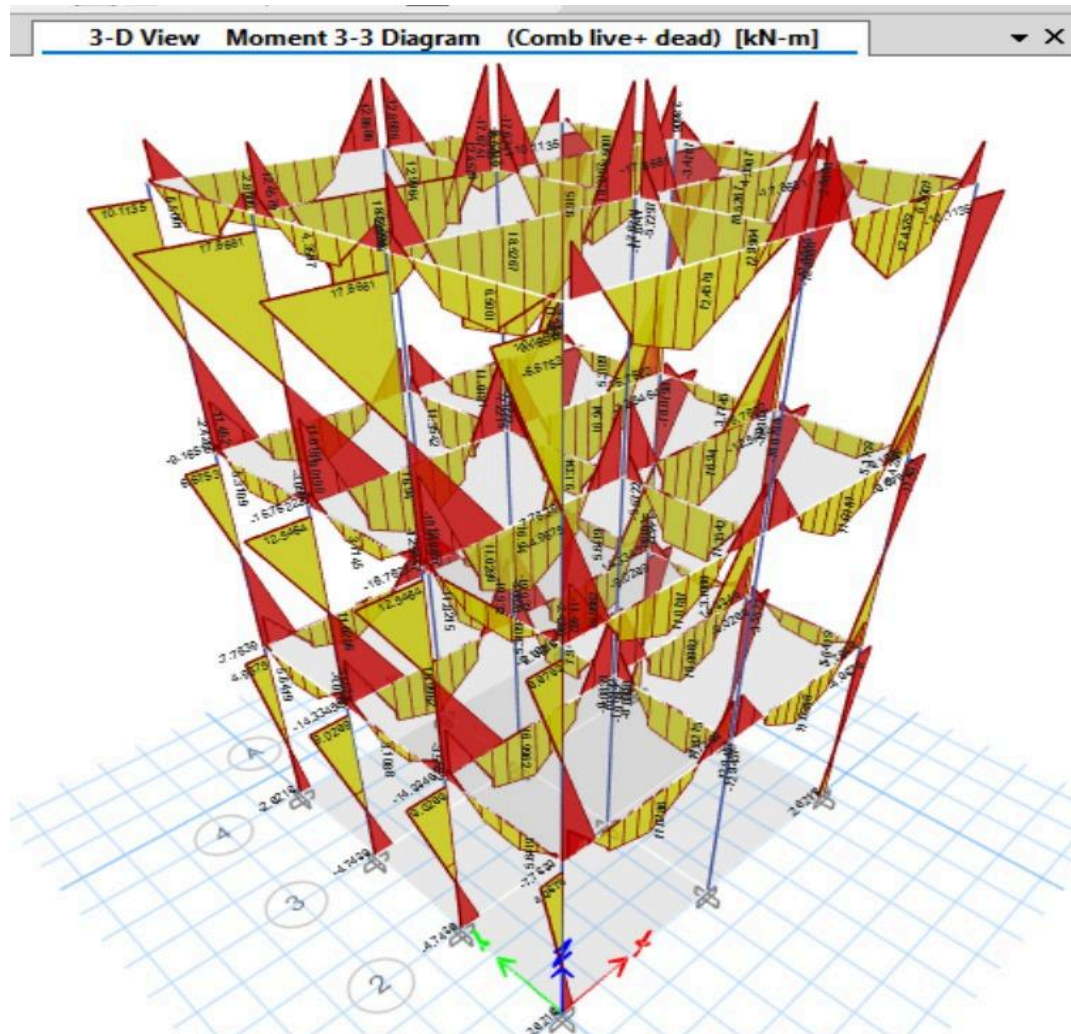
■ 6. Analysis Results — Bending Moment Diagrams (BMD)

The following bending moment diagrams are extracted from ETABS for the governing load combination COMB 1 (1.5(DL+LL)). Results are shown for the critical frame in each direction and for both the RC and Steel models.

◆ 6.1 BMD

(BMD RC)



(BMD steel)

Expected behavior: Beams will show hogging (negative) moments at supports and sagging (positive) moments near midspan. Columns will develop moments at both ends due to frame continuity. The moment at beam-column joints must be in equilibrium.

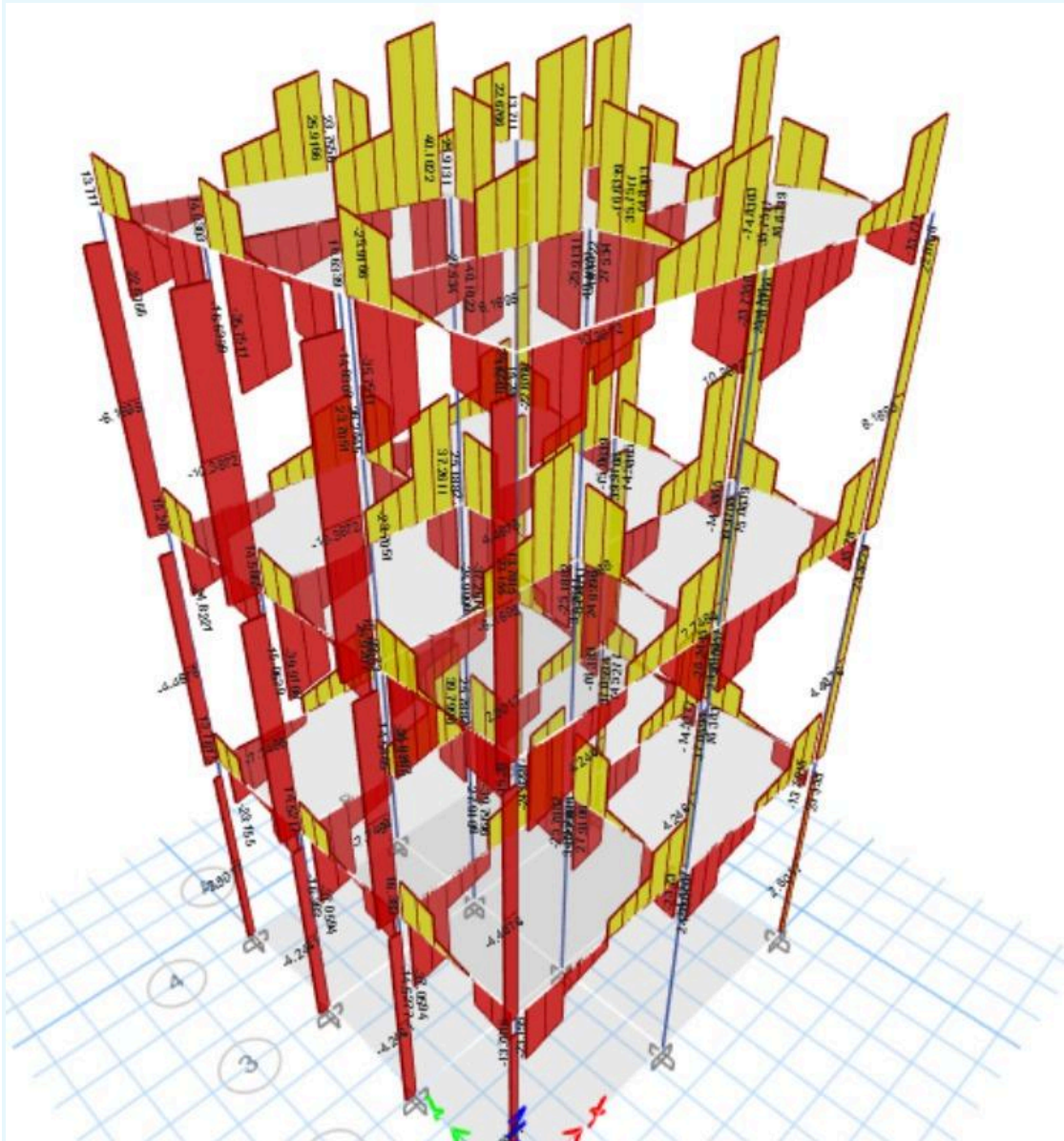
MEMBER	LOCATION	EXPECTED MOMENT (APPROX.)	ETABS VALUE
X-beam (4.5 m), Fl. 1	Mid-span	$w \times 4.5^2 / 16 \approx 47.6$ kN·m*	45.1985 kN·m
X-beam (4.5 m), Fl. 1	Support	$w \times 4.5^2 / 12 \approx 63.5$ kN·m*	47.0913 kN·m

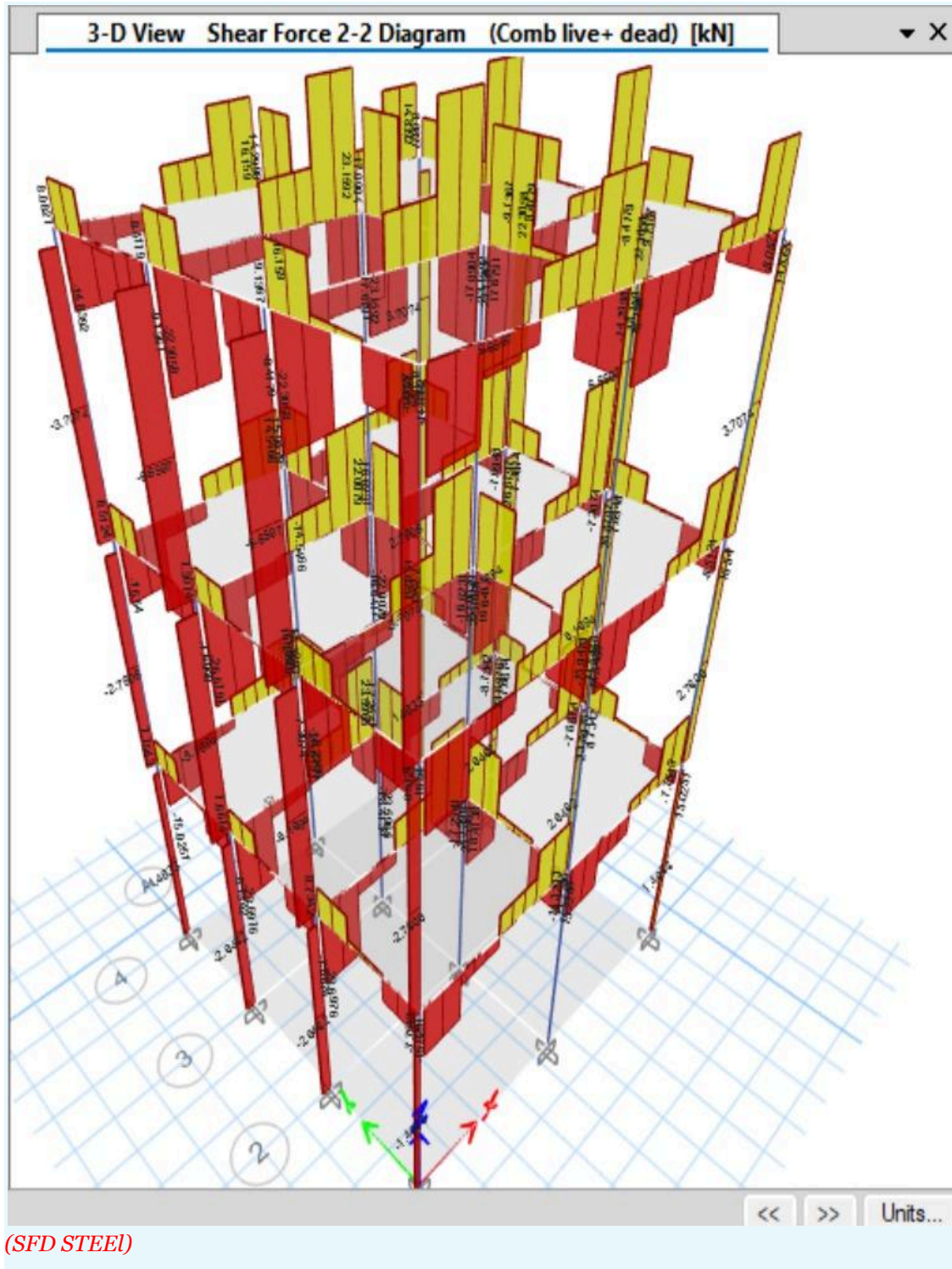
X-beam (4.5 m), Fl. 2	Mid-span	$\approx 47.6 \text{ kN}\cdot\text{m}^*$ (similar)	43.8379 kN·m
Y-beam (3.0 m), Fl. 1	Mid-span	$w \times 3.0^2 / 16 \approx 11.7 \text{ kN}\cdot\text{m}^*$	16.8479 kN·m

■ 7. Analysis Results — Shear Force Diagrams (SFD)

Shear force diagrams are extracted from ETABS for load combination COMB 1. These are used to design shear reinforcement in RC beams and to check the shear capacity of steel beams.

(SFD RC)





◆ 7.1 Maximum Shear Force Summary

MEMBER	LOCATION	EXPECTED SHEAR (APPROX.)	ETABS VALUE
X-beam interior (4.5 m)	At support	$37.6 \times 4.5 / 2 = 84.7 \text{ kN}^*$	71.3358kN
X-beam exterior (4.5 m)	At support	$49.3 \times 4.5 / 2 = 110.8 \text{ kN}^*$	91.3814 kN
Y-beam interior (3.0 m)	At support	$20.8 \times 3.0 / 2 = 31.1 \text{ kN}^*$	41.0396 kN

◆ 7.2 Interpretation

- Maximum shear in X-direction beams occurs at the beam ends (at column faces) for both gravity load cases.
- In continuous beams, the shear at interior supports may be asymmetric due to unequal spans or unsymmetric loading.
- Column shear forces in a gravity-only analysis are small but non-zero due to frame continuity moments.
- These SFD values directly feed into shear stirrup design (Deliverable 2).

■ 8. Analysis Results — Deflection / Deformed Shape

Deflection checks are performed under the serviceability combination COMB 2 (1.0 DL + 1.0 LL) as per IS 456 Cl. 23.2 and IS 800 Cl. 5.6.1. Maximum permitted deflection for beams is the lesser of $L/250$ or 20 mm

◆ 8.1 Deflection Limits — IS 456 / IS 800

MEMBER	SPAN (M)	PERMISSIBLE LIMIT	LIMIT VALUE (MM)
X-beam (Floor 1)	4.5	$L/250$	$4500/250 = 18.0 \text{ mm}$
X-beam (Floor 2)	4.5	$L/250$	18.0 mm
Y-beam (Floor 1)	3.0	$L/250$	$3000/250 = 12.0 \text{ mm}$

Max column drift	5.5	H/500	$5500/500 = 11.0 \text{ mm}$
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◆ 8.2 Observations on Deflected Shape

- Under gravity loads, all beams should deflect downward (positive vertical displacement at mid-span).
- The maximum deflection is expected in the floor beams at mid-span of the 4.5 m X-direction bays (longest span).
- Columns should remain essentially vertical under pure gravity loading with negligible lateral sway.
- The steel frame model is expected to show larger deflections than the RC model due to lower moment of inertia of steel sections vs. RC members.

■ 9. Loads on Sample Columns and Beams

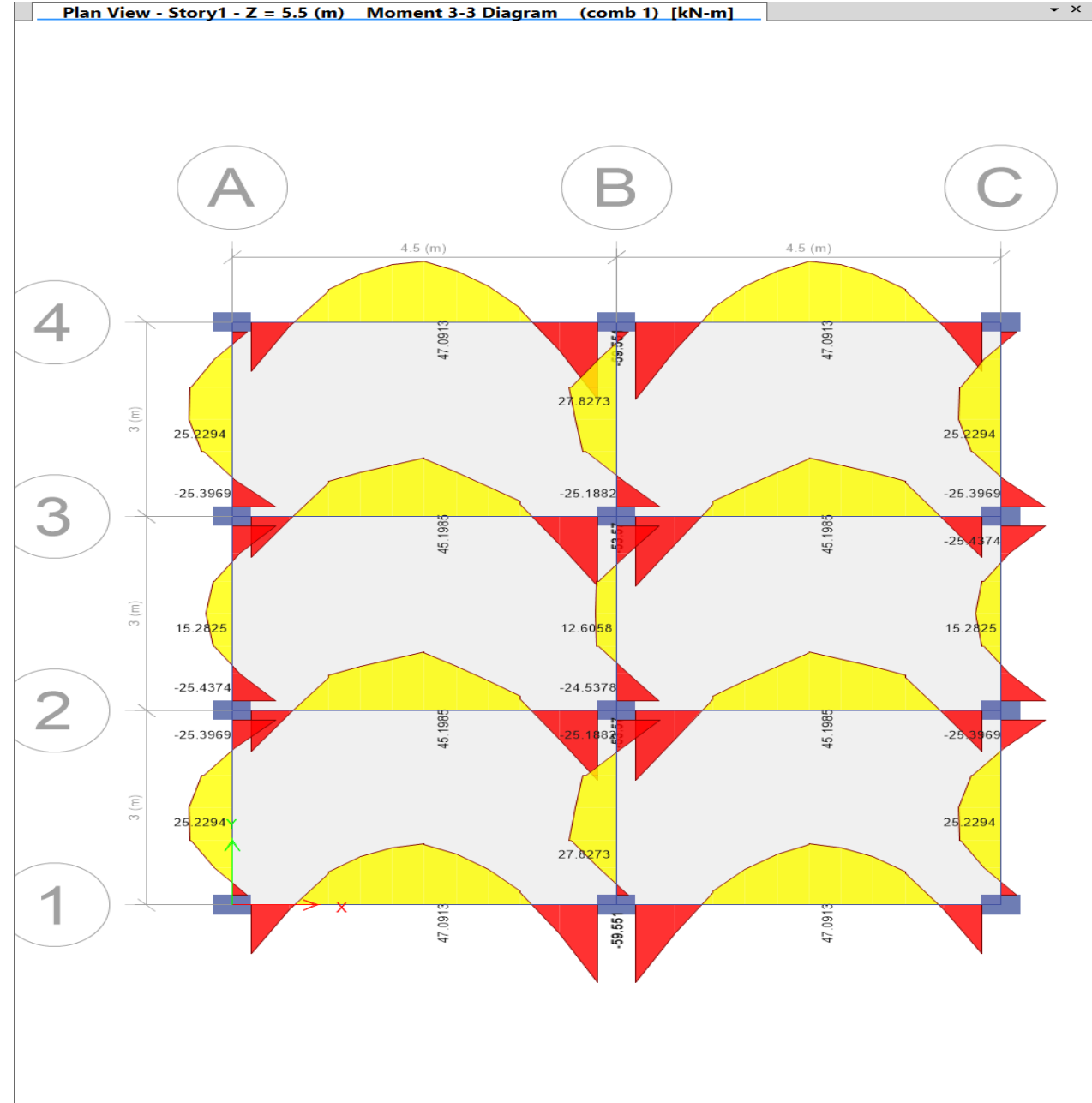
This section presents the axial forces, bending moments, and shear forces extracted from ETABS for selected representative members. These members are the same ones that will be designed manually in Deliverable 2.

◆ 9.1 Member Selection for Manual Design

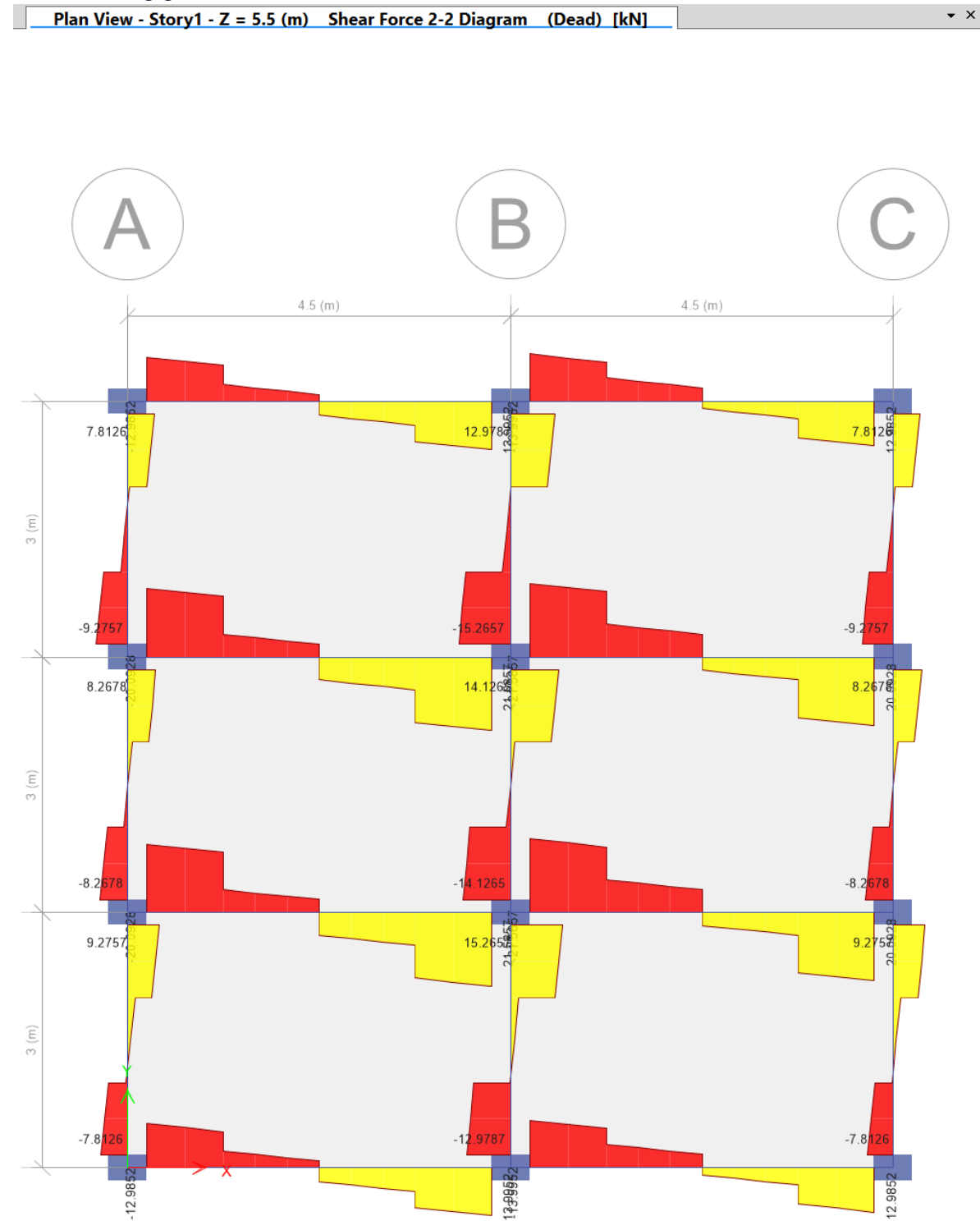
MEMBER TAG	DESCRIPTION	GRID LOCATION	WHY SELECTED
BM-X-int	Interior X-beam	Grid Y=B, Floor 1	Highest loaded X-beam (interior, full tributary width)
BM-X-ext	Exterior X-beam	Grid Y=A, Floor 1	Carries wall load + slab — different loading pattern
BM-Y-int	Interior Y-beam	Grid X=2, Floor 1	Highest loaded Y-beam (interior)
BM-Y-ext	Exterior Y-beam	Grid X=1, Floor 1	Exterior beam, carries wall load
COL-int	Interior column	C-B2, all floors	Maximum axial load — critical for column design
COL-edge	Edge column (Y-edge)	C-A2, all floors	Moderate axial load + higher moment eccentricity

◆ 9.2 Beam Force Summary (ETABS — COMB 1) RC

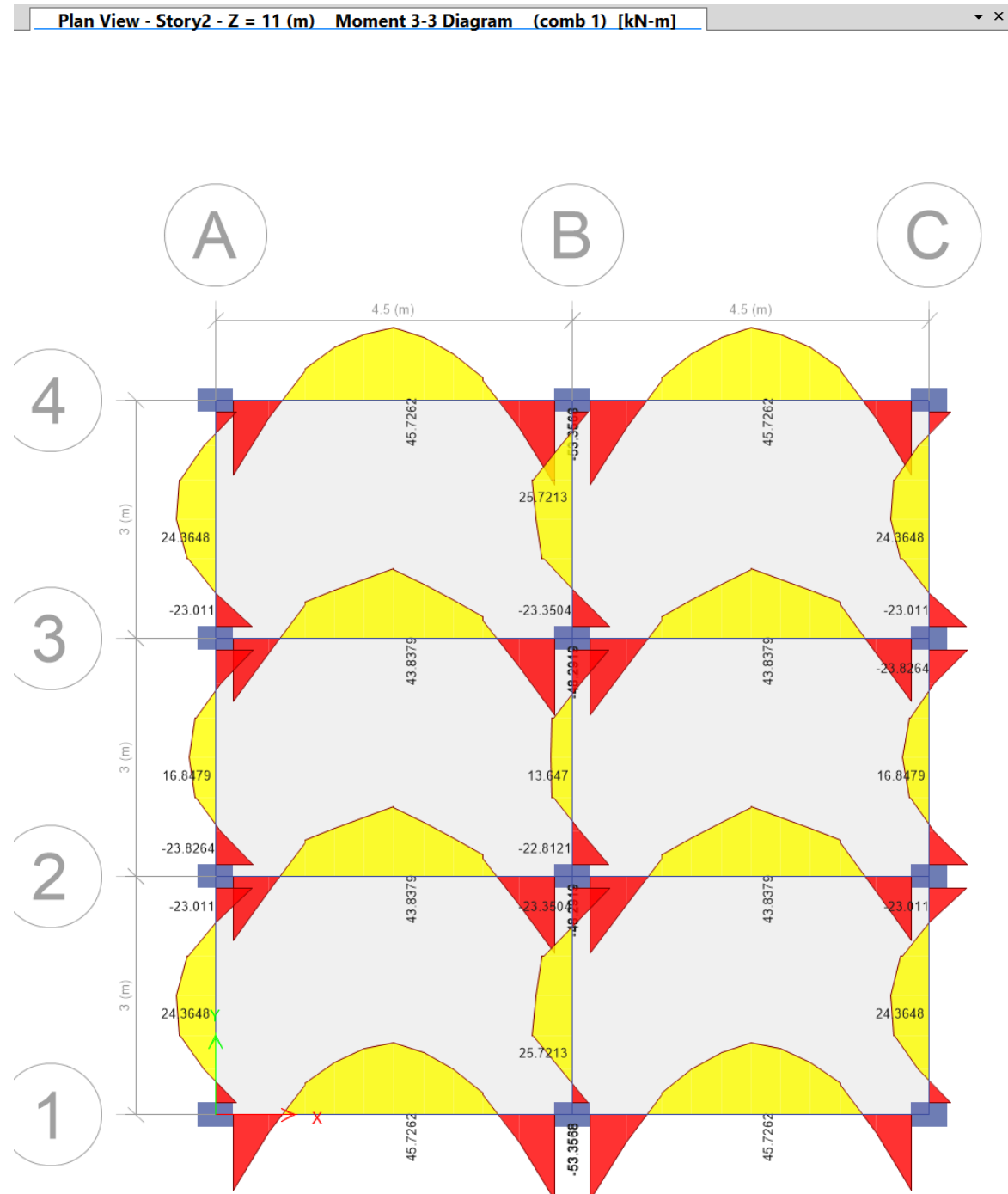
FLOOR 1 Z=5.5M MOMENT



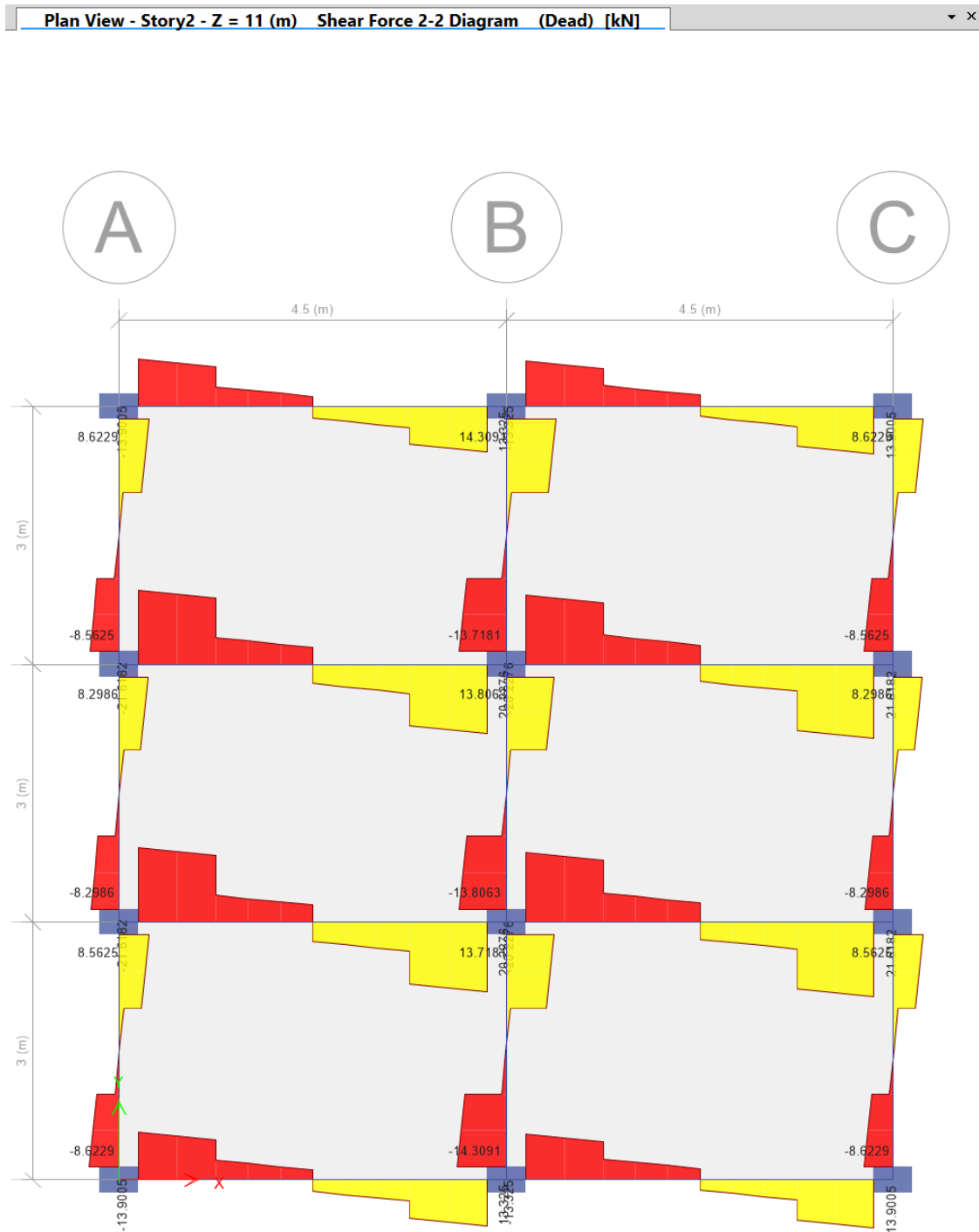
FLOOR 1 Z=5.5M SHEAR DIAGRAM



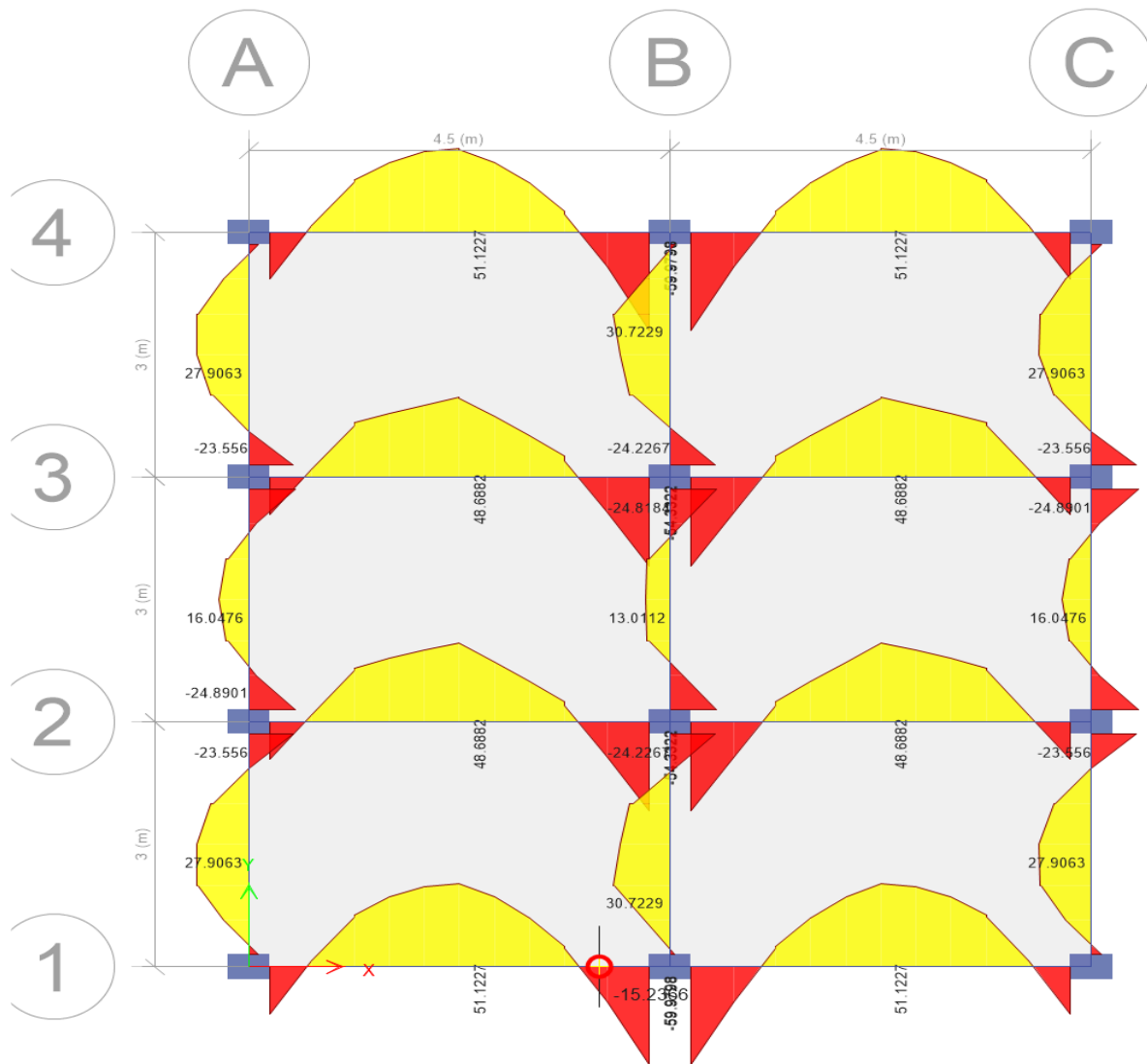
FLOOR 2 Z=11M MOMENT

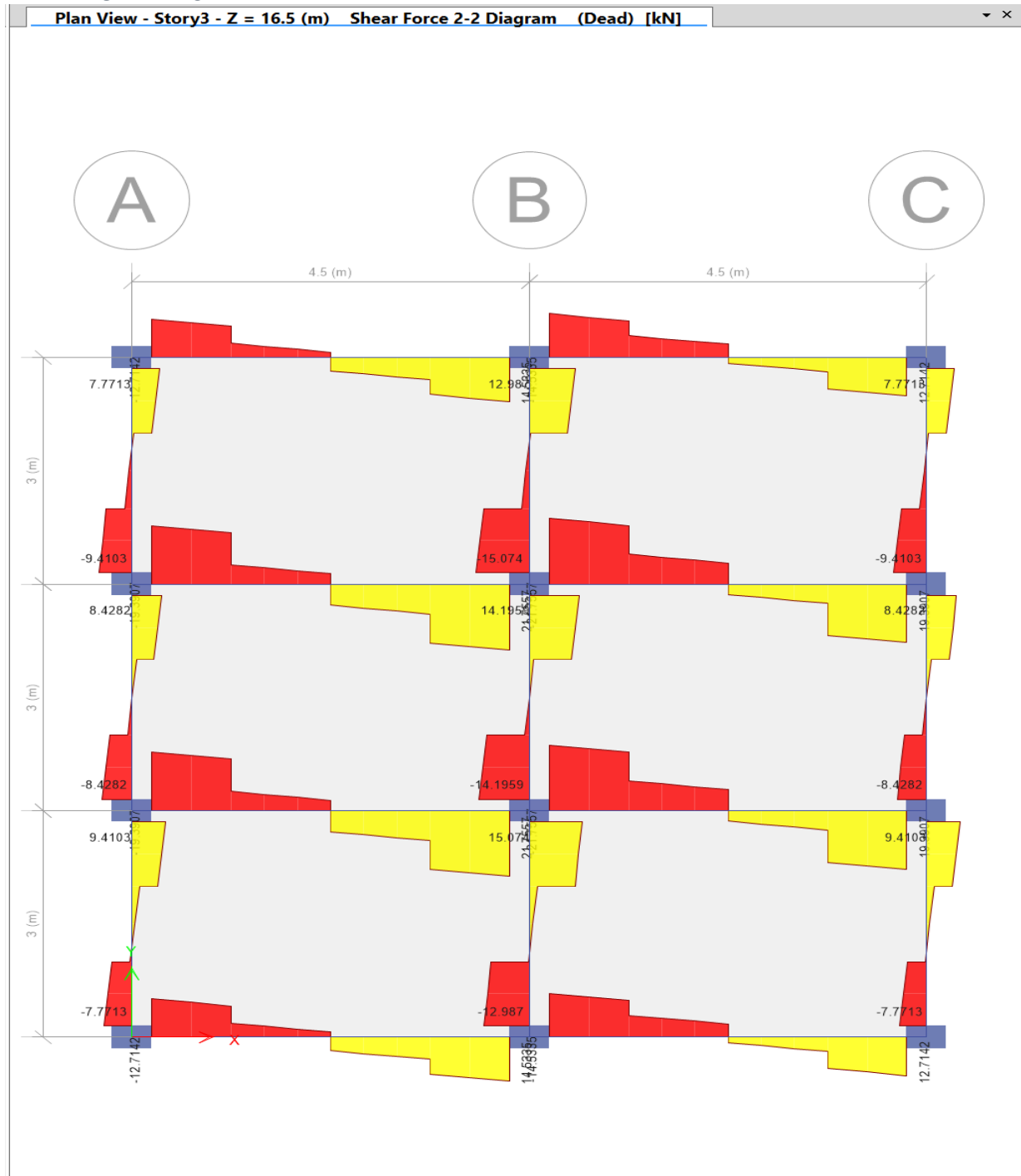


FLOOR 2 Z=11M SHEAR DIAGRAM



FLOOR 3 Z=16.5M MOMENT



FLOOR 3 Z=16.5 SHEAR DIAGRAM

■ 10. Support Reaction Verification

As a critical quality-check step, the sum of all vertical support reactions at the base of the structure must equal the total applied gravity load. Any significant discrepancy indicates modeling errors such as missed loads, incorrect load factors, or improper support conditions.

◆ 10.1 Total Applied Load Estimate

Based on Section 5.4 and including member self-weights (estimated below):

LOAD COMPONENT	CALCULATION	VALUE (kN)
1.5 × Floor DL (2 floors)	$1.5 \times 5.5 \times 81 \times 2$	1,336.5
Roof DL	6.0×81	486.0
1.5 × Wall DL (3 floors)	$1.5 \times 19 \times 36 \times 3$	3,078.0
1.5 × Roof parapet DL	$1.5 \times 4.6 \times 36$	248.4
1.5 × Beam self-weight (est.)	$1.5 \times 51 \times 0.23 \times 0.45 \times 25 \times \text{avg}$	≈ 480
1.5 × Column self-weight (est.)	$1.5 \times 12 \times 0.3 \times 0.45 \times 25 \times 16.5$	≈ 1,005
1.5 × Floor LL (2 floors)	$1.5 \times 2.0 \times 81 \times 2$	486.0
Roof LL	1.5×81	121.5
TOTAL FACTORED 1.5(DL+LL+SW)		≈ 7,545 kN
CHECK (all items already factored)		≈ 7,545 kN

◆ 10.2 ETABS Support Reactions — COMB 1: 1.5(DL + LL)

TABLE: Base Reactions												
Output Case	Case Type	Step Type	Step Number	FX kN	FY kN	FZ kN	MX kN-m	MY kN-m	MZ kN-m	X m	Y m	Z m
Dead	LinStatic			0	0	2808.6977	12639.1395	-12639.1395	0	0	0	0
Live	LinStatic			0	0	648	2916	-2916	0	0	0	0
Modal	LinModEigen	Mode	1	0	-0.7027	0	8.5447	0	-3.1624	0	0	0
Modal	LinModEigen	Mode	2	1.0616	0	0	0	13.1231	-4.777	0	0	0
Modal	LinModEigen	Mode	3	0	0	0	0	0	-5.5525	0	0	0
Modal	LinModEigen	Mode	4	0	-1.8329	0	-5.6465	0	-8.2479	0	0	0
Modal	LinModEigen	Mode	5	-3.4703	0	0	0	5.3254	15.6162	0	0	0
Modal	LinModEigen	Mode	6	0	0	0	0	0	15.7429	0	0	0
Modal	LinModEigen	Mode	7	0	1.7352	0	-6.1302	0	7.8082	0	0	0
Modal	LinModEigen	Mode	8	0	0	0	0	0	17.6884	0	0	0
Modal	LinModEigen	Mode	9	4.5156	0	0	0	14.5338	-20.3204	0	0	0
Modal	LinModEigen	Mode	10	-0.000001072	0.3828	0	-0.3393	0	1.7228	0	0	0
Modal	LinModEigen	Mode	11	-0.00001134	0.5516	0	-0.8113	0	2.4824	0	0	0
Modal	LinModEigen	Mode	12	0.5249	0.00002263	0	0	0.8861	-2.3617	0	0	0
SDL1	LinStatic			0	0	486	2187	-2187	0	0	0	0
RoofDead	LinStatic			0	0	729	3280.5	-3280.5	0	0	0	0
RoofLive	LinStatic			0	0	486	2187	-2187	0	0	0	0
WallDL	LinStatic			0	0	2052	9234	-9234	0	0	0	0
comb 1	Combination			0	0	10814.5465	48665.4593	-48665.4593	0	0	0	0
					total FZ -	7209.6977						

◆ 10.3 Verification Check

CHECK ITEM	MANUAL ESTIMATE	ETABS SUM	DIFFERENCE (%)
Total vertical reaction — COMB 1: 1.5(DL+LL)	≈ 7,545 kN	[7209.6977] kN	[4.44] %
Total DL reaction (unfactored check)	≈ 5,030 kN (÷1.5)	[4806.465] kN	[4.44] %
Horizontal reactions (gravity only — should be ≈ 0)	0 kN	[0] kN	—

◆ 10.4 Tributary Area Check — Interior Column (C-B2)

As an additional verification, the vertical reaction at the interior column C-B2 can be estimated by tributary area and compared to ETABS:

- Tributary area = $(4.5/2 + 4.5/2) \times (3.0/2 + 3.0/2) = 4.5 \times 3.0 = 13.5 \text{ m}^2$
- Factored DL per floor = $1.5 \times 5.5 \times 13.5 = 111.4 \text{ kN}$; Factored LL per floor = $1.5 \times 2.0 \times 13.5 = 40.5 \text{ kN}$
- 3 floors total (factored): DL = 334.1 kN, LL = 121.5 kN → Total = 455.6 kN (approx., excl. SW and wall contributions)
- ETABS reaction at C-B2 under COMB 1 should be in the range of 500–700 kN (factored, including SW and frame redistribution).

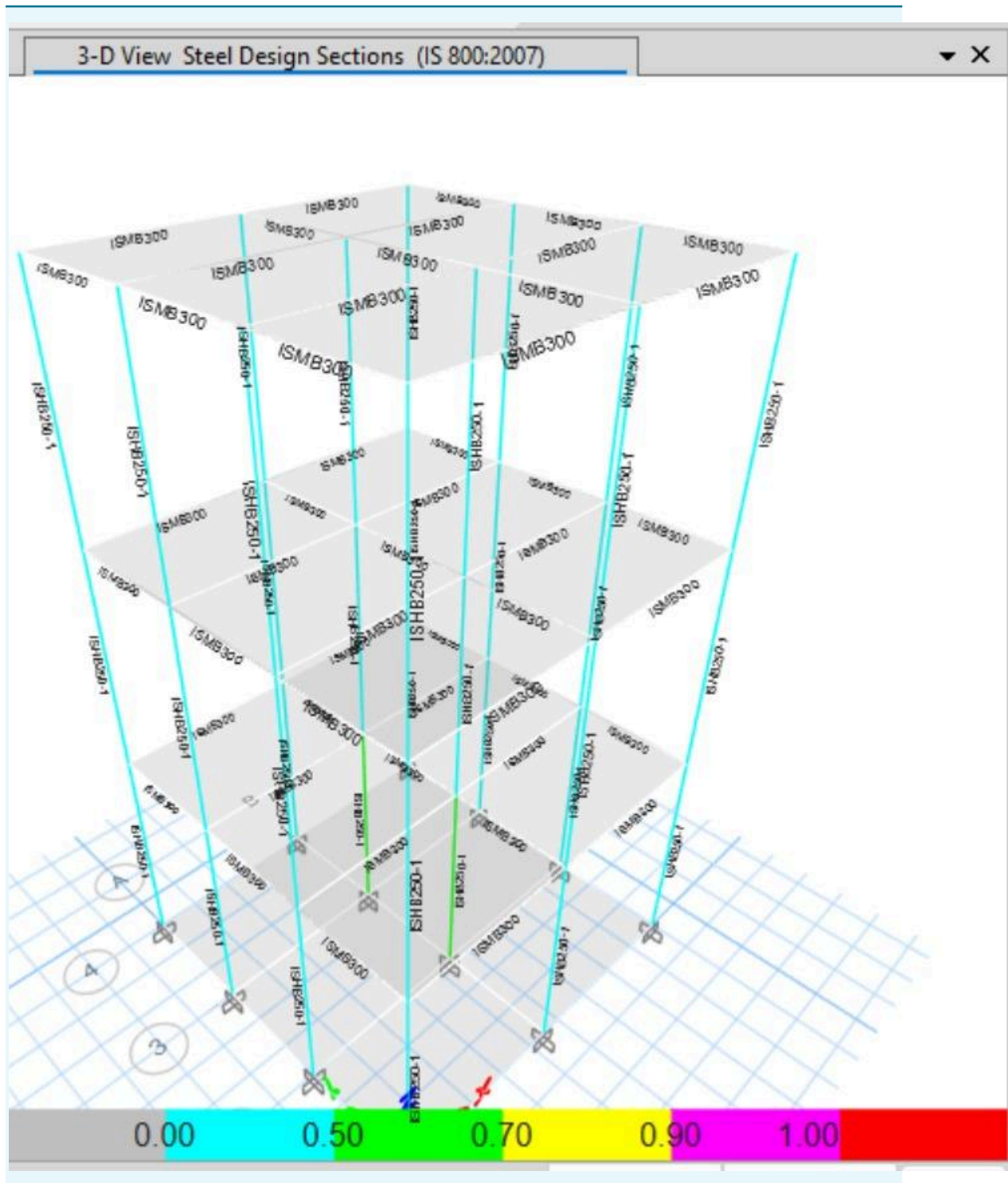
■ 11. Software Models — ETABS

Two separate ETABS v21 models have been built: one for the RC frame and one for the Steel frame. Both models share identical geometry, support conditions, and loading. The following describes key modeling decisions.

◆ 11.1 Modeling Details

MODELING ITEM	RC FRAME MODEL	STEEL FRAME MODEL
Software version	ETABS v21	ETABS v21
Column sections	Rectangular RC 300×450	ISHB 250
Beam sections	RC 230×450 (all directions)	ISMB 300 (all directions)
Slab	Shell element, 150 mm	Shell element, 150 mm
Floor diaphragm	Rigid diaphragm	Rigid diaphragm
Support conditions	Fixed (all 12 columns)	Fixed (all 12 columns)
Self-weight multiplier	1.0 in 'Dead' load case	1.0 in 'Dead' load case
Wall loads	Line load 19 kN/m on perimeter beams	Same
Analysis type	Linear elastic static	Linear elastic static
Steel Design Code	—	IS 800:2007
Concrete Design Code	IS 456:2000	—

◆ 11.3 Steel Frame Model



MATERIAL SECTION PROPERTIES

Frame Section Property Data

General Data

Property Name: ISMB300

Material: Fe345

Display Color: Change...

Notes: Modify/Show Notes...

Shape

Section Shape: Steel I/Wide Flange

Section Property Source

Source: Indian Convert To User Defined

Section Dimensions

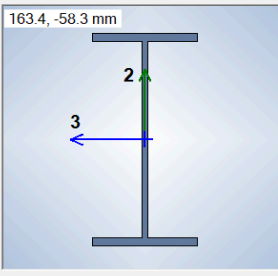
Total Depth	300	mm
Top Flange Width	140	mm
Top Flange Thickness	12.4	mm
Web Thickness	7.5	mm
Bottom Flange Width	140	mm
Bottom Flange Thickness	12.4	mm
Fillet Radius	0	mm

Show Section Properties...

Property Modifiers

Modify/Show Modifiers...
Currently Default

OK
Cancel



Frame Section Property Data

General Data

Property Name: ISHB250-1

Material: Fe345

Display Color: Change...

Notes: Modify/Show Notes...

Shape

Section Shape: Steel I/Wide Flange

Section Property Source

Source: Indian Convert To User Defined

Section Dimensions

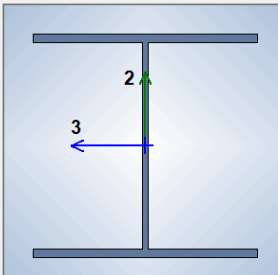
Total Depth	250	mm
Top Flange Width	250	mm
Top Flange Thickness	9.7	mm [Length]
Web Thickness	6.9	mm
Bottom Flange Width	250	mm
Bottom Flange Thickness	9.7	mm
Fillet Radius	0	mm

Show Section Properties...

Property Modifiers

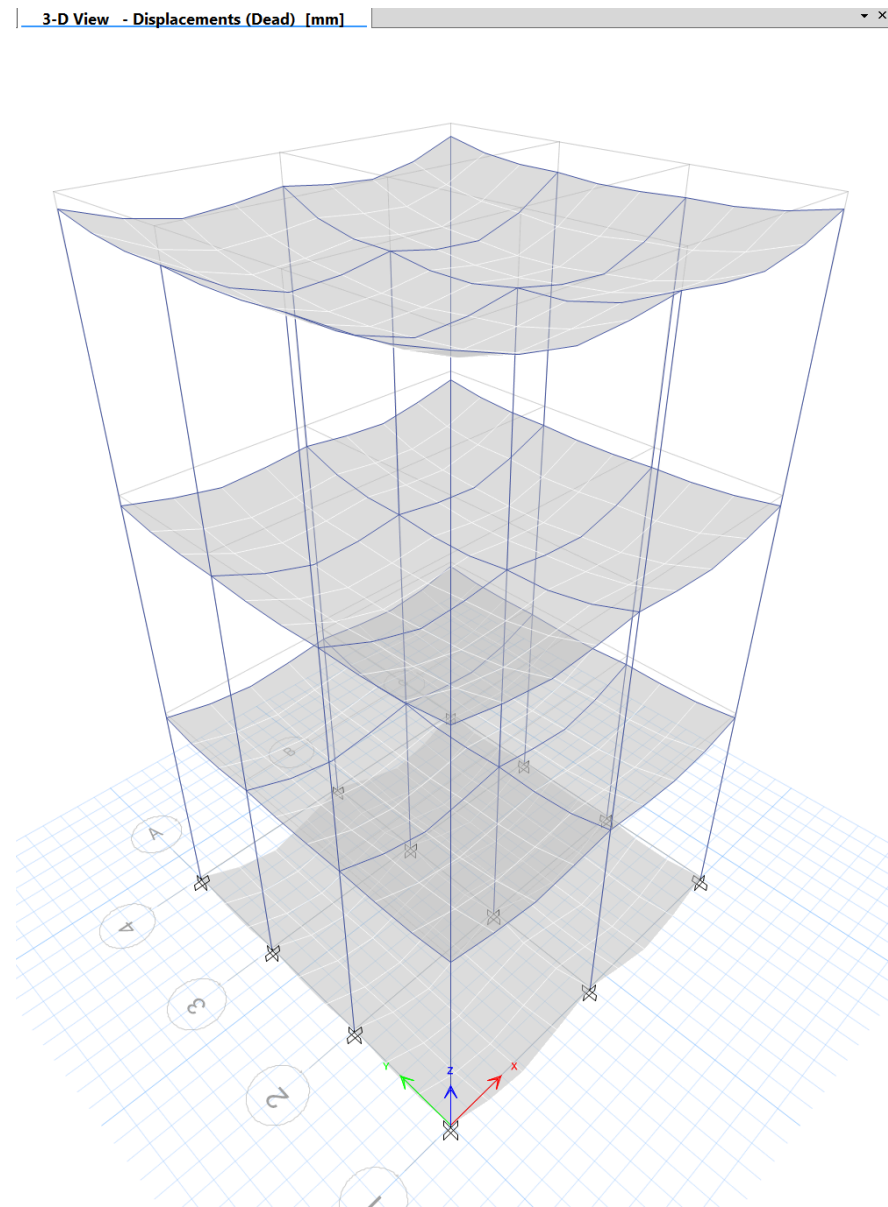
Modify/Show Modifiers...
Currently Default

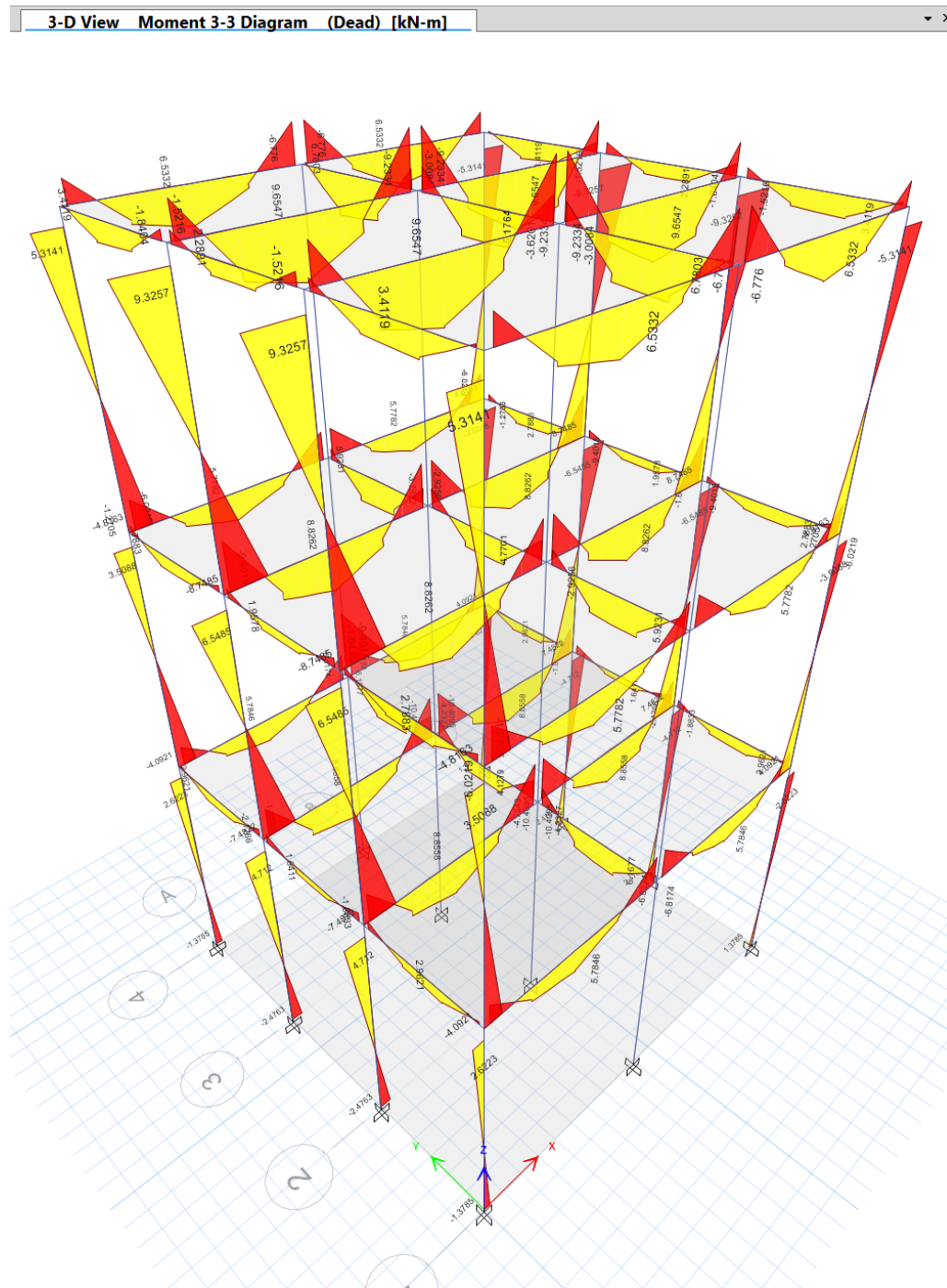
OK
Cancel

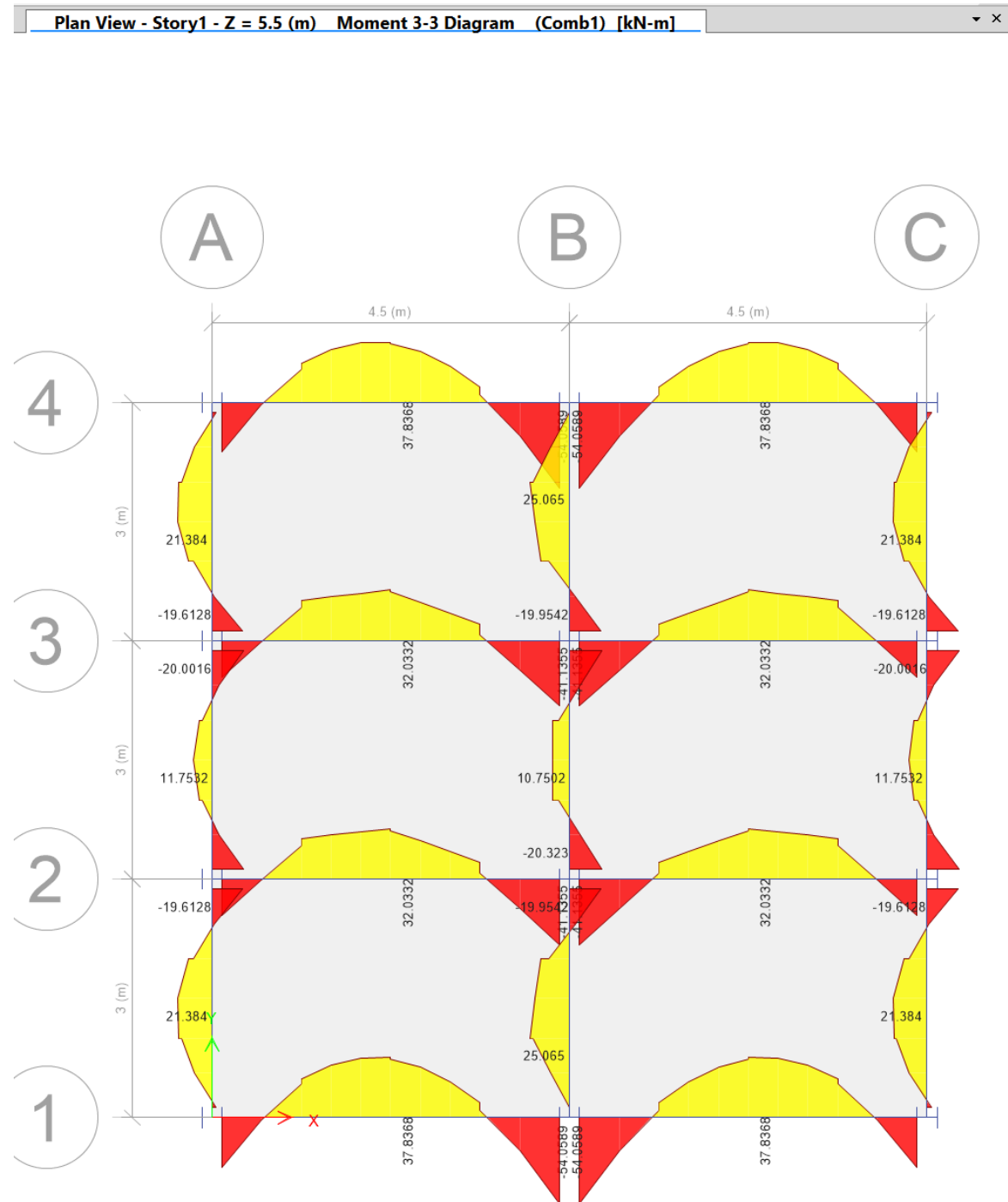


Beam Force Summary (ETABS — COMB 1) STEEL

DISPLACEMENTS

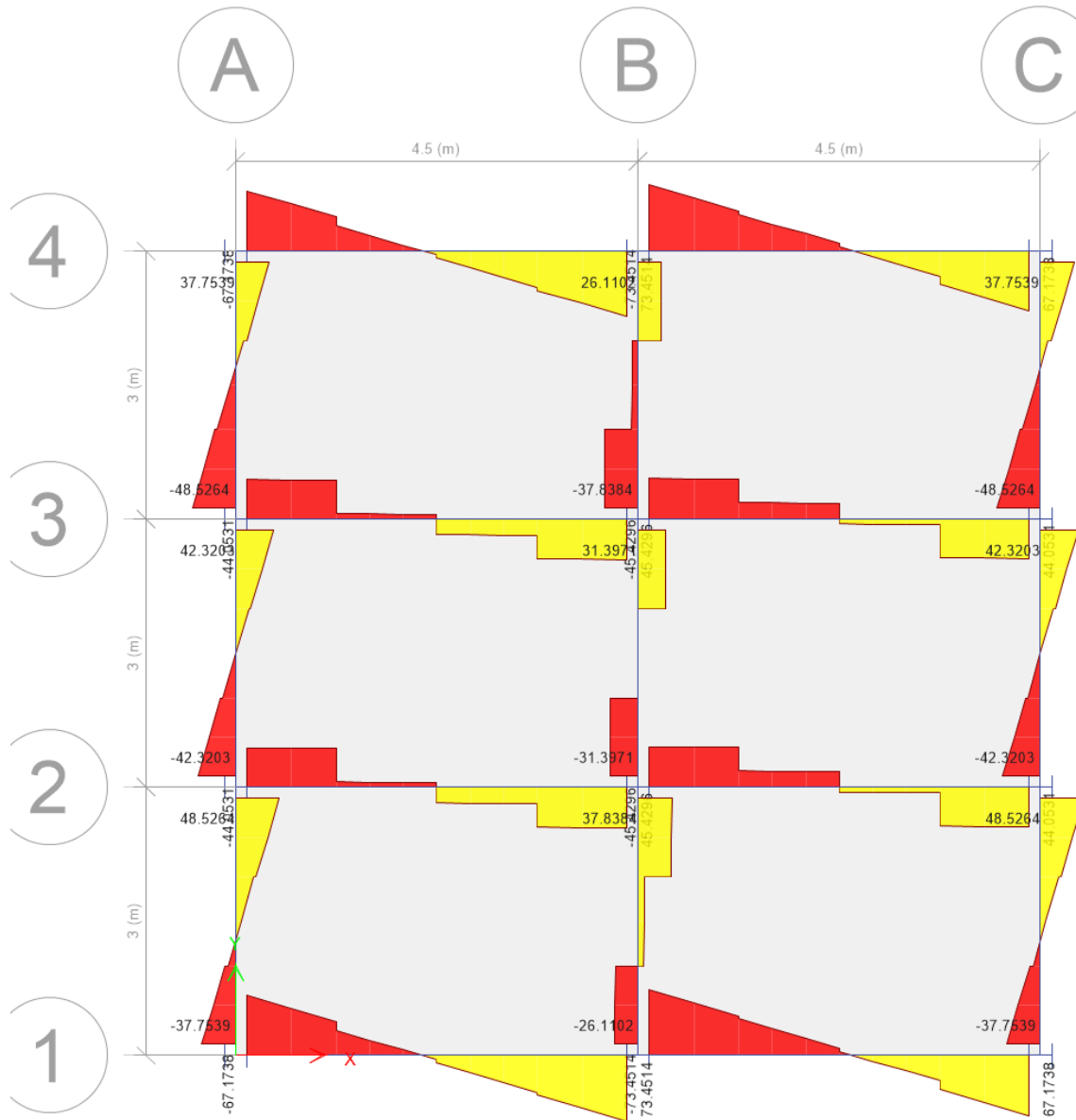




FLOOR 1 Z=5.5M BMD

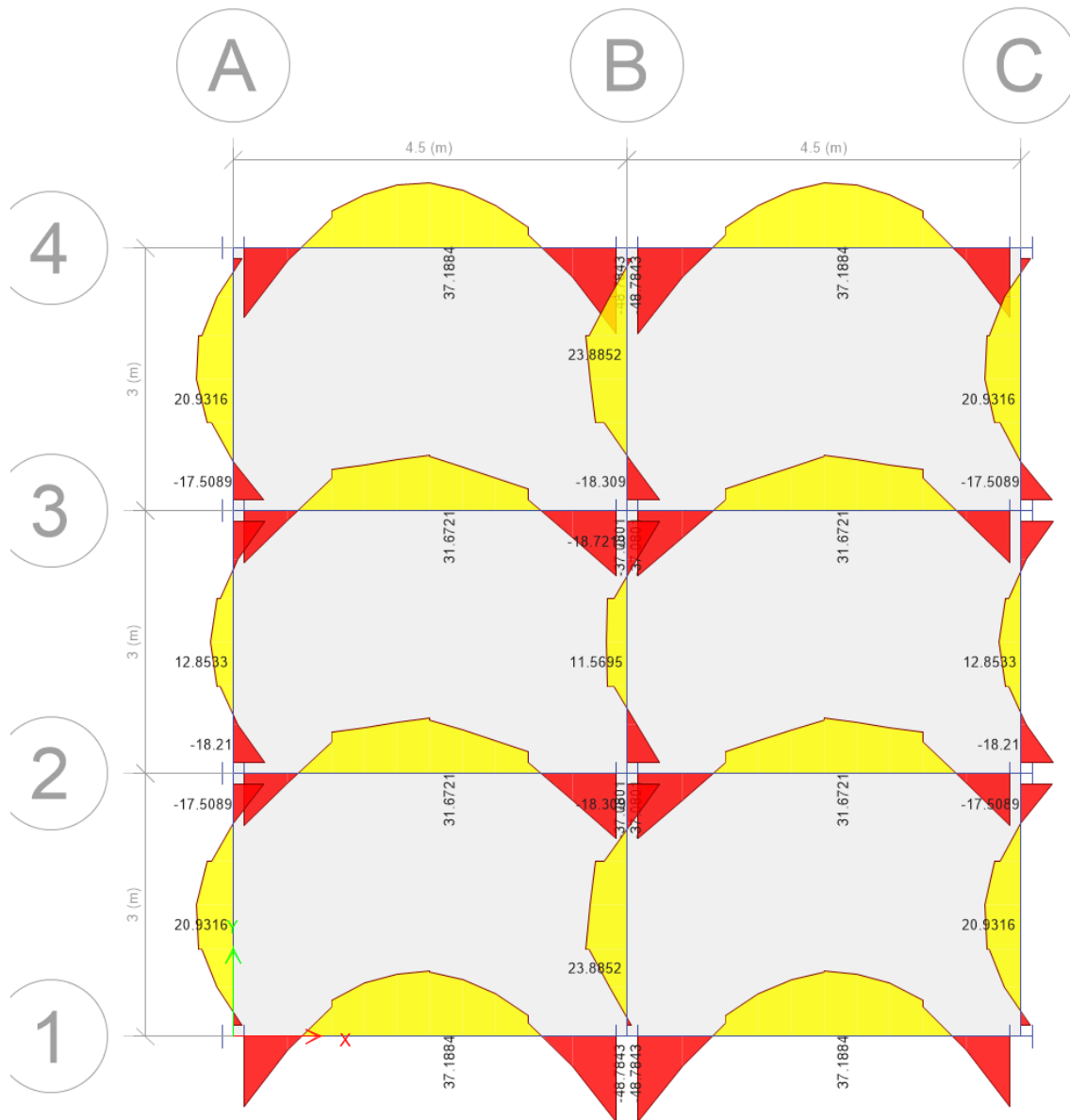
FLOOR 1 Z=5.5M SFD

Plan View - Story1 - Z = 5.5 (m) Shear Force 2-2 Diagram (Comb1) [kN]

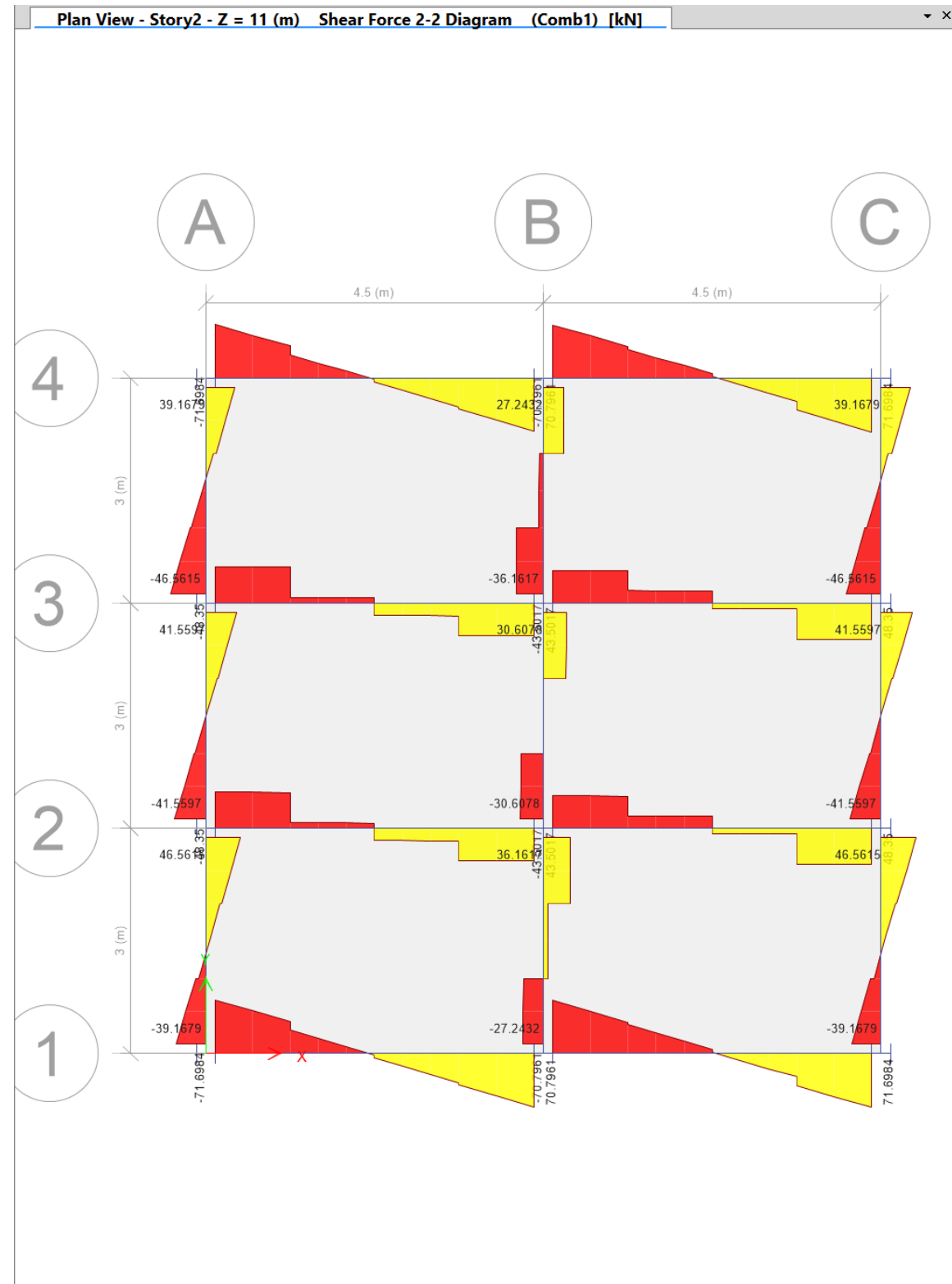


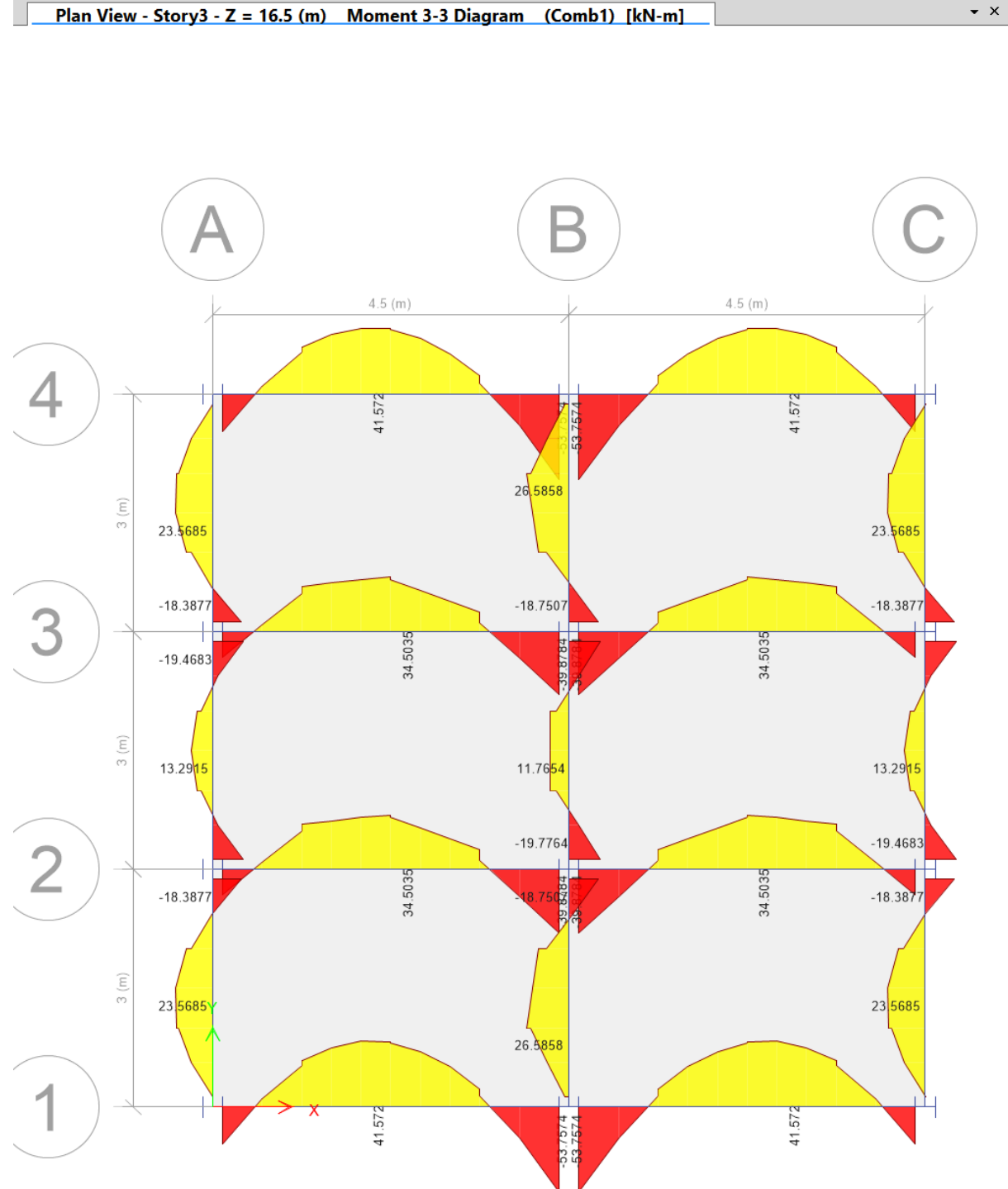
FLOOR 2 Z=11M BMD

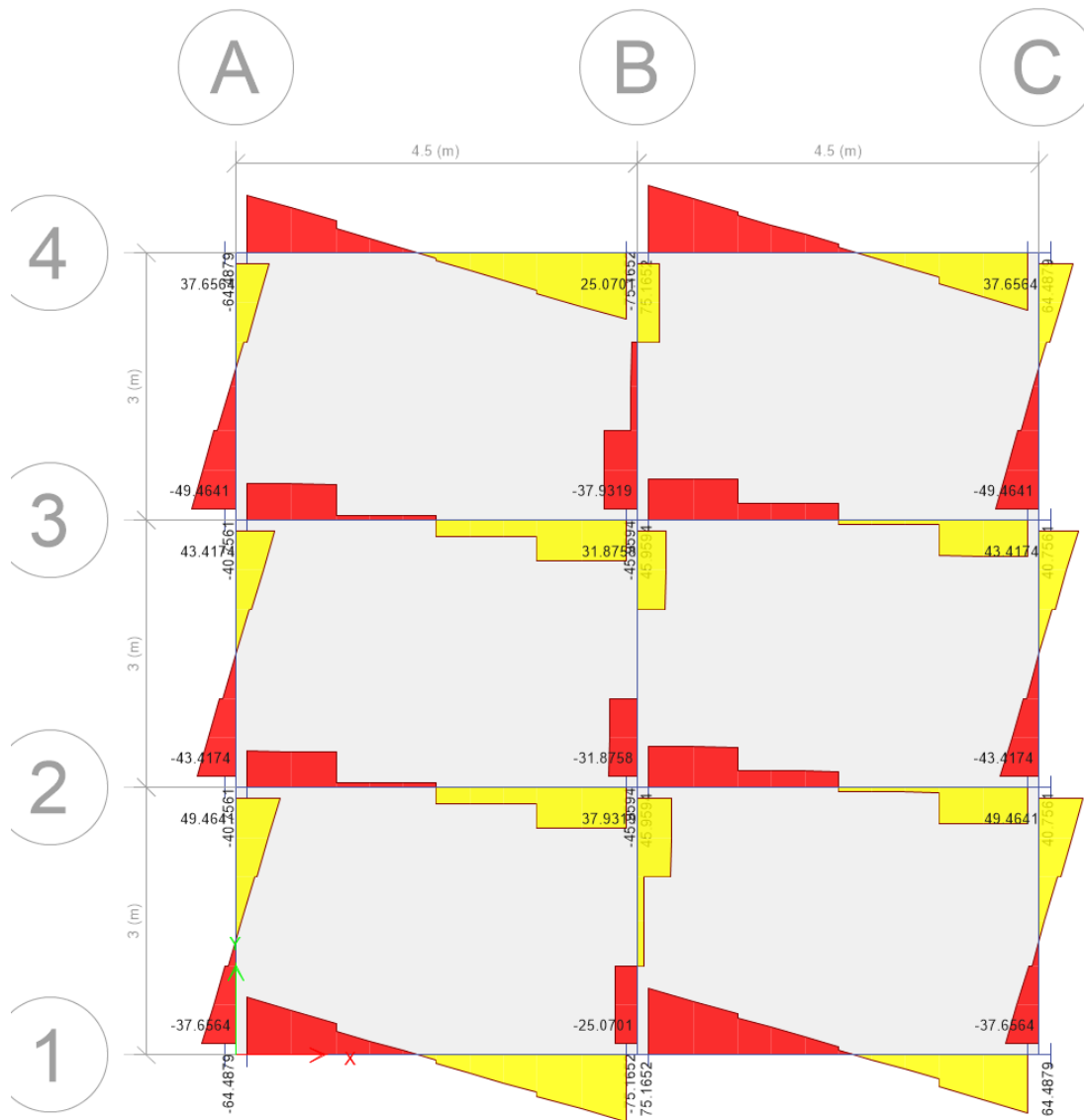
Plan View - Story2 - Z = 11 (m) Moment 3-3 Diagram (Comb1) [kN-m]



FLOOR 2 Z=11M SFD



FLOOR 3 Z=16.5M BMD

FLOOR 3 Z=16.5M SFD**Plan View - Story3 - Z = 16.5 (m) Shear Force 2-2 Diagram (Comb1) [kN]**

■ 12. Design Methodology (for Deliverables 2 & 3)

◆ 12.1 RC Design Approach

- Flexural design of beams: Limit state design per IS 456:2000, singly or doubly reinforced sections.
- Shear design of beams: Design of vertical stirrups per IS 456 Cl. 40. Minimum shear reinforcement per Cl. 26.5.1.6.
- Column design: IS 456 Cl. 39 — combined axial load and uniaxial/biaxial bending. Interaction diagram approach with minimum eccentricity per Cl. 25.4.
- Minimum and maximum reinforcement ratios will be checked per IS 456 Cl. 26.5.
- Deflection and crack width checks per IS 456 Cl. 23.2 and Cl. 35.3.

◆ 12.2 Steel Design Approach

- Beam design: IS 800:2007 Cl. 8 — laterally supported beams; plastic section classification; bending capacity $M_d = \beta_b \times Z_p \times f_y / \gamma_{m0}$.
- Shear design: IS 800:2007 Cl. 8.4 — design shear capacity $V_d = A_v \times f_y / (\sqrt{3} \times \gamma_{m0})$.
- Deflection check: IS 800:2007 Cl. 5.6.1 — $L/250$ limit.
- Column design: IS 800:2007 Cl. 9 — combined axial compression and bending; interaction equations for Class 1/2 sections.

◆ 12.3 Members for Manual Design (per Project Specification)

MEMBER TYPE	DIRECTION	FLOORS	RC DESIGN	STEEL DESIGN
Interior beam	X	All floors	Flexure + Shear	Section + Deflection
Interior beam	Y	All floors	Flexure + Shear	Section + Deflection
Exterior beam	Either	All floors	Flexure + Shear	Section + Deflection

Interior column	—	Each floor	Axial + Bending	Axial + Bending
Edge column	Either	Each floor	Axial + Bending	Axial + Bending

■ 13. Summary of Preliminary Design Decisions

PARAMETER	RC FRAME	STEEL FRAME
Concrete grade	M25 ($f_{ck} = 25 \text{ N/mm}^2$)	M25 (slab only)
Reinforcement/Steel	Fe 500 ($f_y = 500 \text{ N/mm}^2$)	Fe 250 IS 2062
X-dir Beams (4.5 m)	230 × 450 mm	ISMB 300
Y-dir Beams (3.0 m)	230 × 450 mm	ISMB 300
All Columns	300 × 450 mm	ISHB 250
Slab Thickness	150 mm	150 mm RC
Storey Height	5.5 m (all floors)	5.5 m (all floors)
Base Condition	Fixed	Fixed
Floor DL (applied)	5.5 kN/m ²	5.5 kN/m ²
Floor LL (applied)	2.0 kN/m ² (residential)	2.0 kN/m ²
Roof LL	1.5 kN/m ²	1.5 kN/m ²
Wall line load	19 kN/m perimeter beams	19 kN/m perimeter beams
Design combo	1.5(DL+LL) — IS 875 Pt. 5	1.5(DL+LL) — IS 875 Pt. 5
RC design code	IS 456:2000	—
Steel design code	—	IS 800:2007

■ 14. References

- IS 456:2000 — Plain and Reinforced Concrete — Code of Practice (4th revision). Bureau of Indian Standards, New Delhi.
- IS 800:2007 — General Construction in Steel — Code of Practice (3rd revision). Bureau of Indian Standards, New Delhi.
- IS 875 (Part 1):1987 — Code of Practice for Design Loads (Other Than Earthquake) for Buildings and Structures: Part 1 Dead Loads. BIS, New Delhi.
- IS 875 (Part 2):1987 — Code of Practice for Design Loads: Part 2 Imposed Loads. BIS, New Delhi.
- IS 875 (Part 5):1987 — Code of Practice for Design Loads: Part 5 Special Loads and Combinations. BIS, New Delhi.
- SP 6(1):1964 — Handbook for Structural Engineers: Structural Steel Sections. BIS, New Delhi.
- IS 2062:2011 — Hot Rolled Medium and High Tensile Structural Steel. BIS, New Delhi.
- ETABS v21 — Software Documentation and Help Manual. Computers and Structures Inc., Berkeley, CA.
- Pillai, S.U., and Menon, D. (2009). Reinforced Concrete Design (3rd ed.). Tata McGraw-Hill, New Delhi.
- Subramanian, N. (2011). Steel Structures: Design and Practice. Oxford University Press, New Delhi.

***IF THERE IS ANY CHANGES OR FIXES PLEASE MENTION OR LET US KNOW ***